```
(.venv) PS D:\iot\kafka> python producer.py
 Tool Wear: 83.31
[2026-08-18T16:16:00.688396] Air Temp: 302.88K | Process Temp: 318.39K | RP
 Tool Wear: 264.15
[2026-08-18T16:16:02.705076] Air Temp: 299.62K | Process Temp: 309.85K | RP
| Tool Wear: 16.0
[2026-08-18T16:16:04.814036] Air Temp: 302.65K | Process Temp: 313.71K | RP
| Tool Wear: 3.41
[2026-08-18T16:16:07.013091] Air Temp: 296.0K | Process Temp: 314.46K | RPM
 Tool Wear: 104.87

Sensor simulator stopped.
(.venv) PS D:\iot\kafka> python producer.py
```

```
py4j.protocol.Py4JError: An error occurred while calling o95.awaitTermination
(.venv) PS D:\iot\spark> docker exec -it iot-spark /opt/spark/bin/spark-submit --packages org.apache.s
park:spark-sql-kafka-0-10_2.13:4.1.3 --conf spark.jars.ivy=/tmp/.ivy2 /opt/spark-apps/streaming.py
```

docker exec -it iot-spark /opt/spark/bin/spark-submit --packages org.apache.spark:spark-sql-kafka-0-10_2.13:4.1.3 --conf spark.jars.ivy=/tmp/.ivy2 /opt/spark-apps/streaming.py

```
KeyboardInterrupt
(.venv) PS D:\iot\kafka> python consumer.py
```

```
✓ Container iot-backend    Running
(.venv) PS D:\iot\docker> docker compose build backend
```

docker compose build backend

docker compose ps

docker compose up -d

```
(.venv) PS D:\iot\frontend> npm run dev
  VITE v8.2.1  ready in 3559 ms

  ➜  Local:   http://localhost:5173/
  ➜  Network: use --host to expose
  ➜  press h + enter to show help
(.venv) PS D:\iot\frontend>
```